Vitamins

2025

Aml Almakhzoum

- **Vitamins** are organic substances, different in chemical structures, and functions, needed daily in small amounts (μg—mg), and Many of them act as coenzymes.
- They do not enter in the structure of the tissues or oxidized by them.
- **RDA(recommended dietary allowance**): (quantity of vitamin recommended for daily ingestion to meet essential needs of healthy person).

- Daily requirement varies with age, body size, growth, exercise, pregnancy, lactation, illness.

- To treat deficiency states, vitamins are given 2-3 times RDA, or more

- **Provitamins**: These are precursors of vitamins that converted into vitamins inside the body e.g. carotenes are provitamin

The vitamins are mainly classified into two According to the solubility:

A. **The fat soluble vitamins :**these are vitamins (**A, D, E and K)**

1. They are soluble in fat solvents.
2. They need bile salts for absorption.
3. They can be stored In the body

B. **Water soluble vitamins:** these are vitamins **(B complex and C)**.

1. **They are soluble in water**
2. **Most of them are not stored in the body**.

Sources of vitamins:

- Dietary (no one food contains all vitamins)
- Partially endogenous synthesis (B3, D)
- totally endogenous synthesis (carotenes to retinal)
- Intestinal bacteria (K2, Biotin)
- Pharmaceutical synthesis

1-Water soluble vitamins are needed in daily supply , easily absorbed and easily excreted via urine , so, stored for few weeks, except B12, (3-4 years in liver),

2-Fat soluble vitamins stored for a long time , in liver and adipose tissue, excreted via feces

Absorption

- Their absorption depends on healthy bowel mucosa.
- Some disease such as Alcoholic liver disease , cystic fibrosis, and chronic use of laxatives decrease absorption.

- **It is advised** to take your vitamins with a snack or meal to avoid stomach irritation.
- The presence of carbohydrates and proteins stimulate digestive enzymes that will allow for better absorption of nutrients. (Iron should be taken on an empty stomach)

Deficiency (hypovitaminosis):

causes:

1-Mal nutrition: Inadequate intake of fat diet, and vegetables

2-Mal absorption: biliary obstruction, regional ileitis diseases, and pernicious anemia(lack of intrinsic factor) .

3-Mal metabolism : failure to convert to active form, lack of a transport protein

4-Increased requirement : growth, pregnancy, lactation, wound healing.

5-Increased excretion

6-**Drugs**: antibiotic, and isoniazid for tuberculosis (antagonist of B6)

- Symptoms of deficiency appear after depletion of the stored vitamin
- In **malnutrition** symptoms of different vitamins appear

Hypervitaminosis (toxicity)

Over doses for long period, especially fat soluble vitamins

Elderly persons are more susceptible for toxicity:

1- the percentage of fat increase with age ,

2- deterioration of renal function

Fat soluble vitamins

- Hydrophobic molecules
- Have **isoprene unit**

- Their absorption depends on bile, pancreatic secretion, and healthy bowel mucosa, absorbed through **chylomicron** (a lipoprotein carrier), and transported by a binding protein.

Vitamin A

Chemistry:

i. **Vitamin A** is fat soluble. It is an active form , present only in animal tissues

ii. The pro-vitamin, (**beta-carotene)** is present in

plant tissues.

-It sensitive to oxygen and ultraviolet light

Sources:

1- animal sources :liver, milk, butter, fish.(retinol ester)

2-plant sources: present as a precursor (β –carotene in carrot).

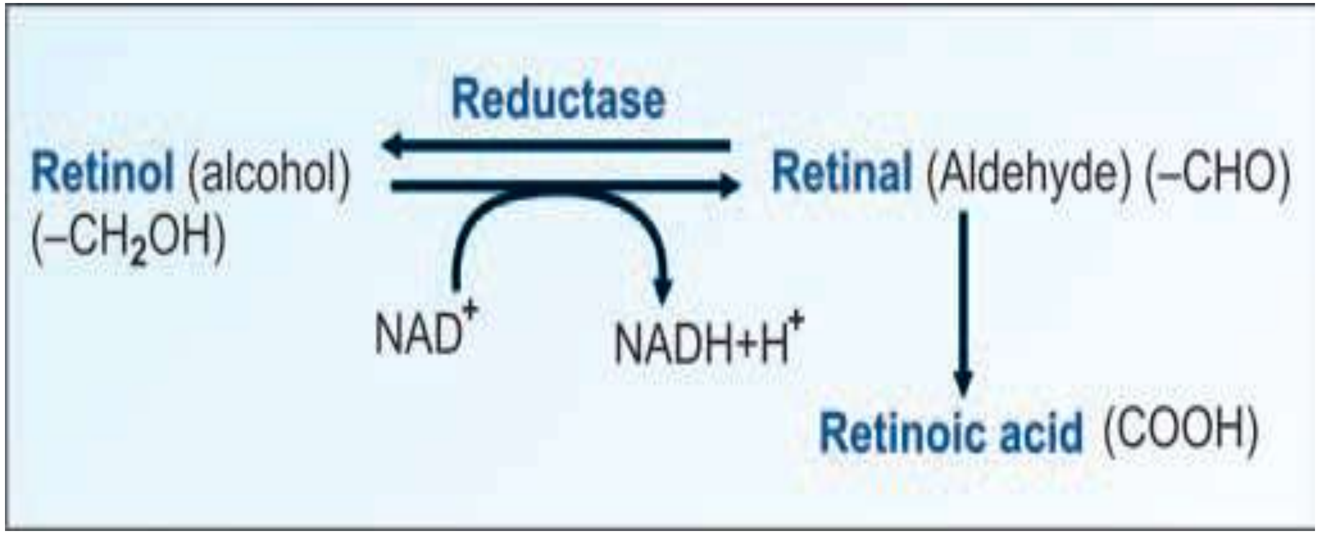
Vitamin A

-One molecule of beta carotene can theoretically give rise to two molecules of vitamin A.
Three different compounds with **vitamin A activity** are:
1- **retinol** (vitamin A alcohol),
2-**Retinal** (vitamin A aldehyde)
3- **retinoic acid** (vitamin A acid), All the compounds are called (**retinoids**).

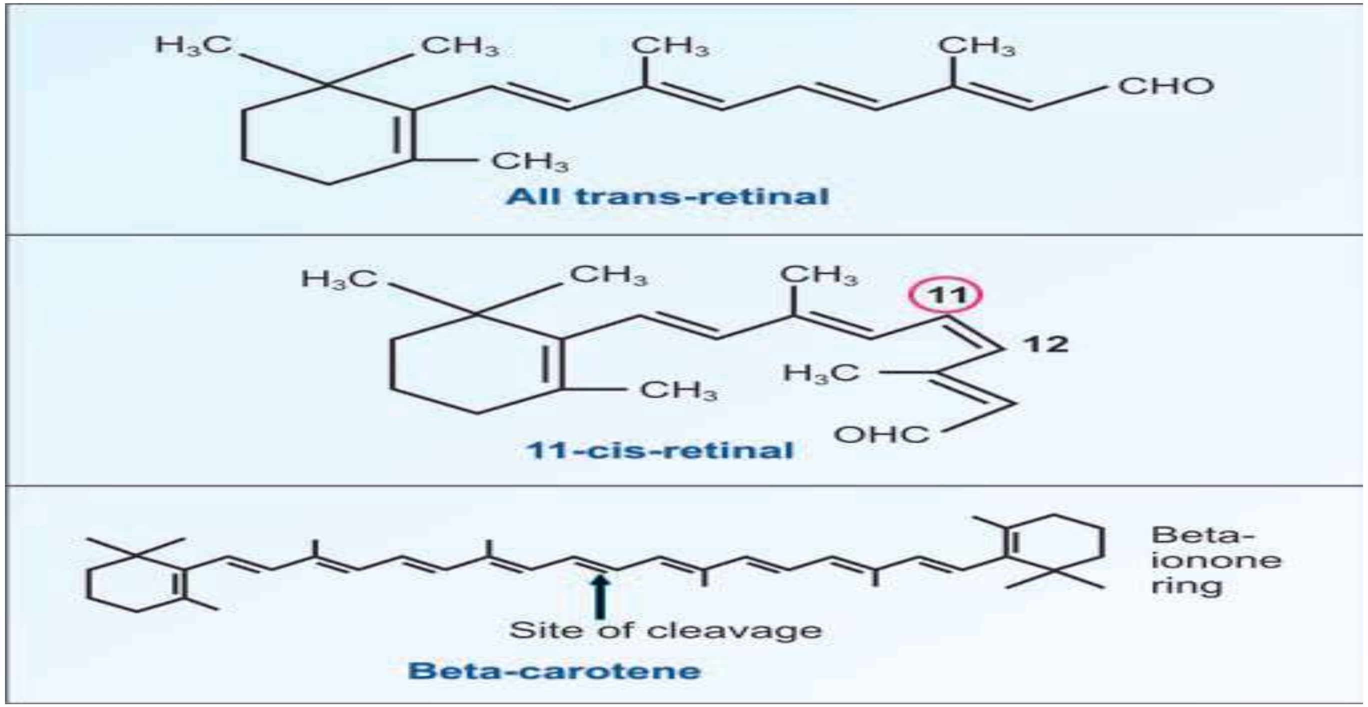
-The retinal may be **reduced** to retinol by retinal reductase. This reaction is readily reversible.
-Retinal is **oxidized** to retinoic acid, which cannot be converted back to the other forms

-The all-trans variety of retinal, also called **vitamin A1** is most common .
-**Vitamin A2** is found in fish oils and has an extra double bond in the ring.
Biologically important compound is 11-cis-retina

Interconversion of vitamin A molecules



Structure of vitamin A



Digestion and absorption of Vitamin A

Diet contains retinol esters and beta-carotene.

1- Retinol esters (animal source) are hydrolyzed into fatty acids and **retinol** that absorbed into intestinal mucosal cells.

2-Beta carotene(plant source) is cleaved by a di-oxygenase, to form retinal. The retinal is reduced to **retinol** by an NADH or NADPH dependent retinal reductase present in the intestinal mucosa.

Absorpation:

Intestine is the major site of absorption

The absorption is along with other fats and requires bile salts

Within the mucosal cell, the **retinol is reesterified with fatty acids**, incorporated into **chylomicrons** and transported to liver.

3-In the liver stellate cells, vitamin is stored as **retinol palmitate**.

Metabolism:

1. When the body cells need vitamin A, stored **retinol esters are hydrolyzed** and free retinol combines with a protein formed by the liver called **retinol binding protein (RBP).** RBP carries retinol to the retina, skin, gonads and other target cells.

a) In retina, **retinol is converted into retinal** that essential for visions.

b**) In other target cells** retinol is oxidized into retinoic acid which binds to nuclear receptors. Retinoic acid receptor complexes stimulate genes. This mode of action is similar to that of hormones

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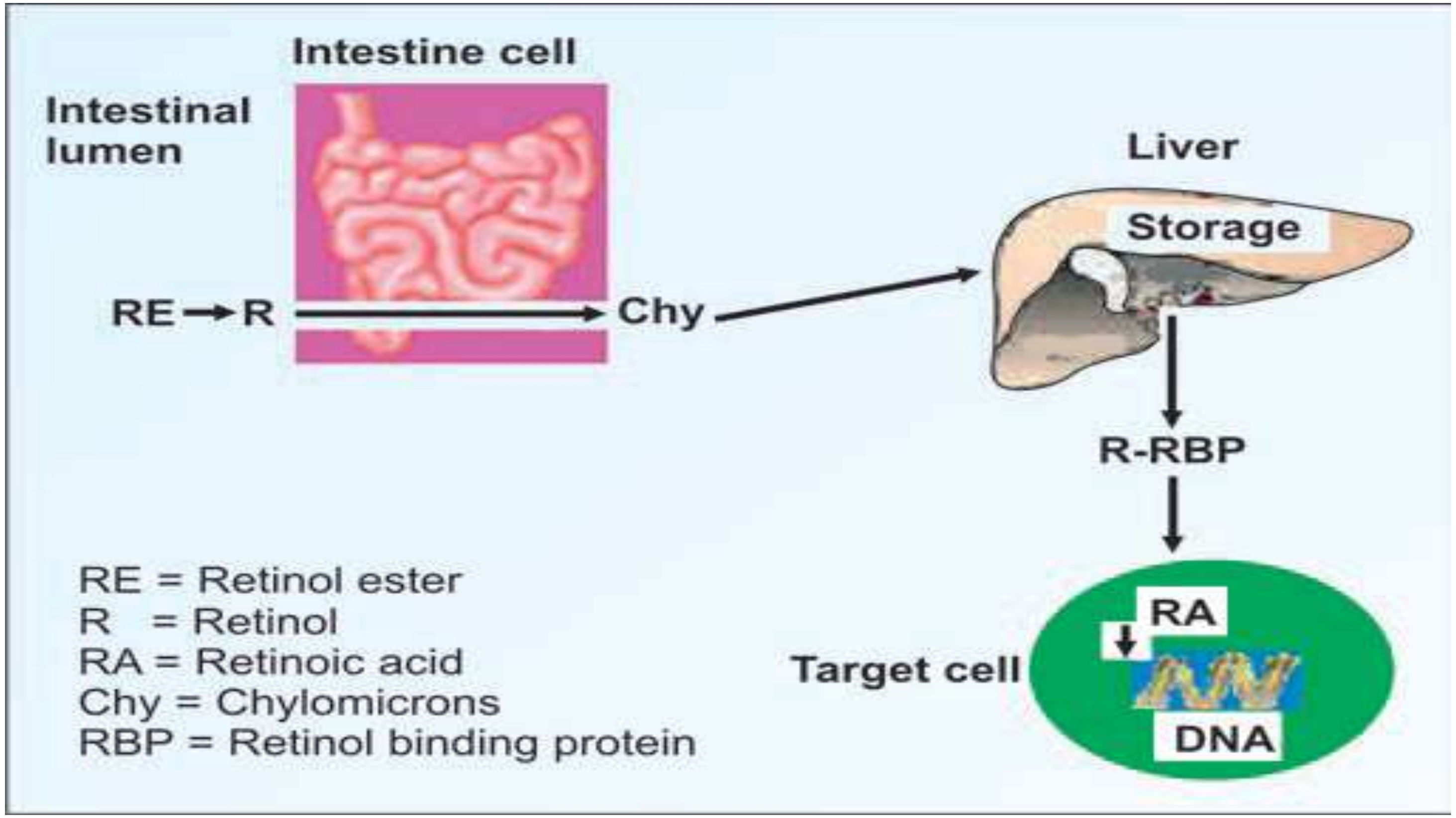
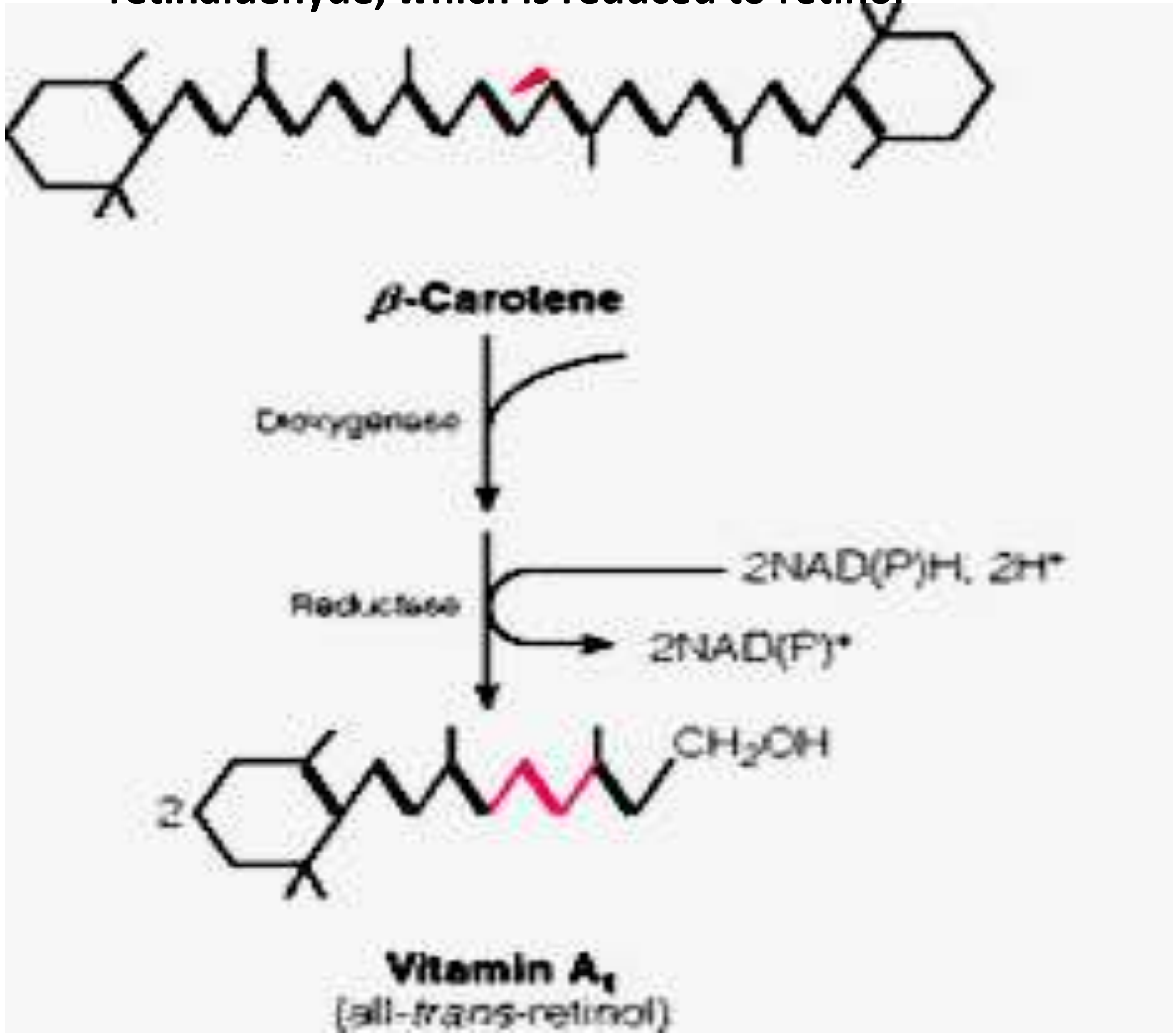
Vitamin A metabolism



Beta-carotene is cleaved by carotene dioxygenase, yielding retinaldehyde, which is reduced to retinol



-**RDA (recommended dietary allowance)**

- 6 µg of β-carotene is equivalent to 1 µg of preformed retinol.

-

Children: 2000-3500 iu,

-Men :5000 iu , wamen:4000 iu

Function of vit A

1-Vision: Generates pigments for the retina (**retinal**)

2-Maintains surface lining of eyes (inner surface of the eyelids which are covered with a membrane called conjunctiva)...(**retinal**).

V.A Has a Role in the Regulation of Gene Expression & Tissue Differentiation

acting Like the steroid hormones:
retinol and retinoic acid bind to nuclear receptors that along with the receptor binds to the response elements of DNA. to produce that exert their function **.**

-**retinoic acid and retinol** regulate growth, development and differentiation of tissues and immune system cells
-**Retinol** functions in development of bone and teeth

-**Retinol** plays a role in male and female reproductive tissues and are essential for fertility.

- **Carotenes** are antioxidants effective at low O2 concentrations
- **Retinoids and carotenes** have anti cancer activity

Vision

The human retina contains two types of receptor cells for vision **cones and rods**:

1-Cones in the retina:- are responsible for vision under bright light and translate objects to color vision
They contain the photosensitive protein (iodopsin), is (cone proteins also, 11-cis-retinal).
Reduction in number of cones or the cone proteins, will lead to color blindness

2-**Rods** in the retina:- are responsible for vision in dim light and translate objects in black and white vision
They contain the photosensitive protein (Rhodopsin) .
Rhodopsin is made up of **11-cis-retinal** + **opsin**.
Deficiency of 11**cis-retinal** will lead to night blindness

Visual cycle

1-When rhodopsin is exposed to dark light, **11- cis retina** is (isomerization)converted into **all trans retinal.**

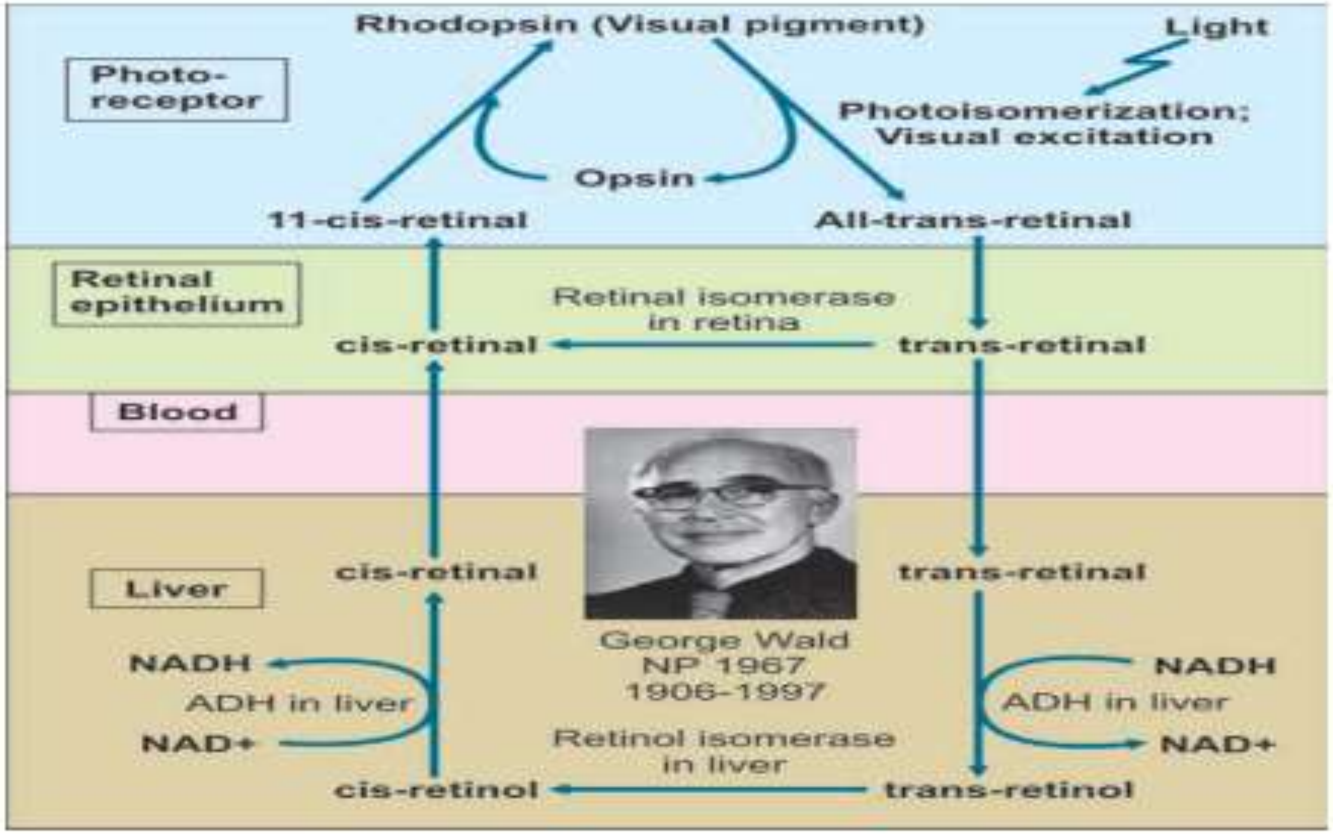
2-The conformational changes of 11-cis retinal to all trans retinal induce changes in permeability to cations, decreasing Na+ and increasing Ca++ ions influx in membranes of rod cells, triggering nerve impulses , allowing light and the image to be perceived by the brain

vision

- For vision, rhodopsin must be regeneraed.

- All trans retinal are converted back to 11-cis retinal.

visual cycle.



Vitamin A deficiency

1-The earliest sign of deficiency is a loss of sensitivity to green light, followed by impairment of adaptation to dim light, Followed by night blindness.

-More prolonged deficiency leads to **xerophthalmia:**

(Dry cornea &conjunctiva, cloudiness & ulceration, keratinization of epithelium, cornea, skin & complet blindness).

2-poor wound healing

3-Reduced reproductive performance

4- increased susceptibility to infectious diseases.

VIT A Toxicity

VA exceeds the capacity of RBP (retinol binding protein)
Leading to
Bone enlargement, muscle pain, loss of appetite, headache, dry itchy skin, Skin coloration, hair loss, Headaches, fatigue, vomiting, increased liver size, and affects the CNS.

-vit A also has **teratogenic effects** in fetus

Vitamin D(calciferol)

Is a steroid hormone.

-Not sensitive to oxygen
-Resistant to alkalies and high temperatures

- **Daily Requirements: 400 IU/day in infant**
  Adult 800-1000 IU/day.

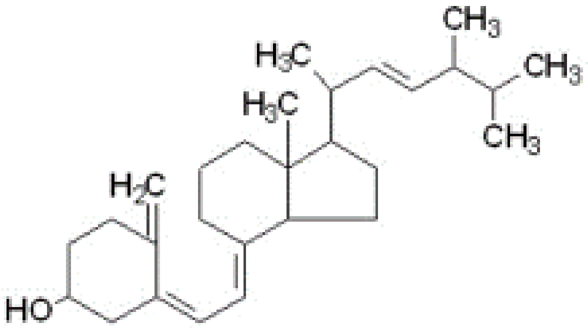
Sources of Vitamin D:

1. Ultraviolet rays of sun generate the vitamins from the provitamin **ergosterol** (in plants) and **7 dehydrocholesterol** in human.

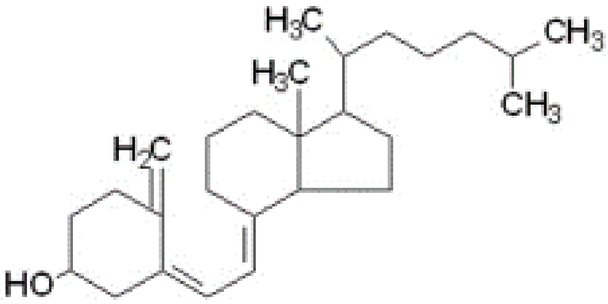
2. Liver, egg, milk and fish liver oils are rich in vitamin D.

Synthesis:

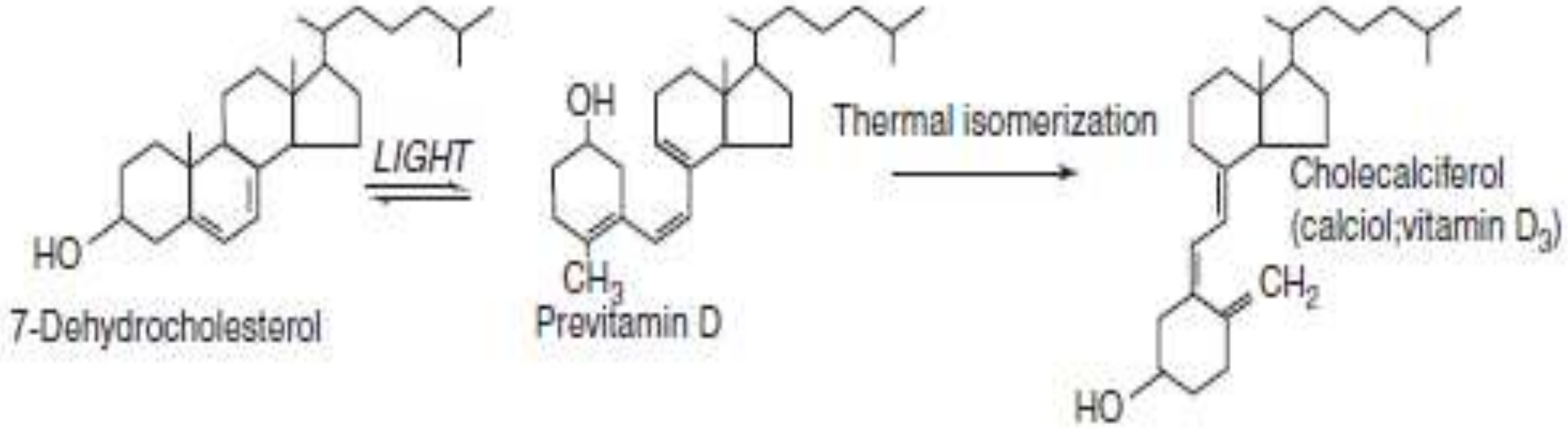
7-Dehydrocholesterol (in the skin) undergoes a nonenzymic reaction on exposure to ultraviolet light, yielding previtamin D, This undergoes a further reaction to form **cholecalciferol(vit D3)**, which is absorbed into the bloodstream.



Ergocalciferol
Vitamin D_2



Cholecalciferol
Vitamin D_3



HO
7-Dehydrocholesterol
LIGHT
OH
CH3
Previtamin D
Thermal isomerization
Cholecalciferol
(calciol;vitamin D3)
CH2
HO

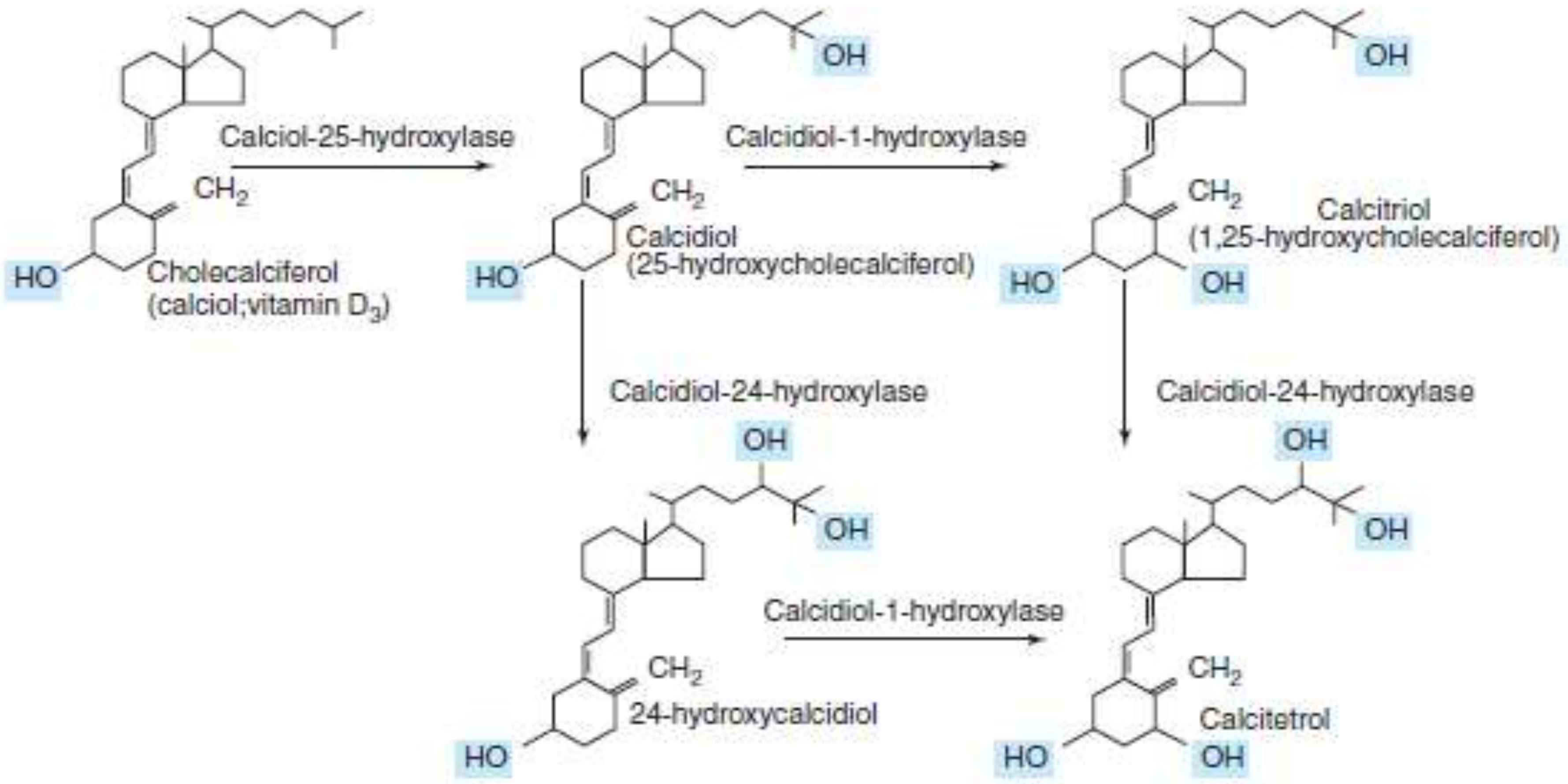
Vit D Metabolism

-Vitamin D Is Metabolized to the Active Metabolite, (**Calcitriol**) in Liver & Kidney

-**Cholecalciferol**, either synthesized in the skin or from food, undergoes two hydroxylations to yield the active metabolite**.**

- After absorption, transported to the liver by α_2-globulin
- **In liver**, cholecalciferol hydroxylated at C-25 to produce 25-hydroxycholecalciferol **(storage form).**
- Then Transported to the **kidney** where hydroxylated at C-1
- So the active form is formed **(1-25 dihydroxy cholecalciferol** ,also named **calcitriol**),
- or **24-hydroxylation** to yield a probably inactive metabolite, **24,25-dihydroxyvitamin D** (24-hydroxycalcidiol), directs vitamin D metabolites toward inactivation and excretion, preventing excess buildup of the active form of vit D.

-These reactions done by **hydroxylase**, Mg, NADPH, O2 , cytochrome 450 (monooxygenase)



Calciol-25-hydroxylase
Calcidiol-1-hydroxylase
Cholecalciferol
(calciol;vitamin D3)
Calcidiol
(25-hydroxycholecalciferol)
Calcitriol
(1,25-hydroxycholecalciferol)
Calcidiol-24-hydroxylase
Calcidiol-24-hydroxylase
Calcidiol-1-hydroxylase
24-hydroxycalcidiol
Calcitetrol
CH2
HO
OH

The **enzyme 1-hydroxylase** in kidney is controlled by:

1-**Parathyroid** hormone increase 1-hydroxylase.

2-**Level of Ca, and P in serum**(low level activate 1-OHase).

normal level of Ca: 9-11mg% , P: 2.5-5 mg%

3-**Feed back inhibition** (increases 1-25DHCC, decreases 1-OHase and increase 24-OHase)

Mode of action

- Calcitriol bind to a receptor in nucleus to induce synthesis of Ca- binding protein, which stimulate:

1-Intestinal absorption of Ca and P

2-Renal reabsorption of Ca and P

3-Mobilization of Ca and P from the bone

-**The main function** of vitamin D is in the control of calcium and phosphate homeostasis.
-also Influences differentiation and proliferation of cells
-involved in insulin secretion
-synthesis and secretion of parathyroid and thyroid Hormones
-activate T and B lymphocytes

-vitamin D is protective against autoimmune, cancers, and cardiovascular diseases, also protective against prediabetes and the metabolic syndrome

- In children (**Rickets**):the bones of children are undermineralized as a result of poor absorption of calcium.

In adults (**Osteomalacia**)results from the demineralization of bone, especially in women who have little exposure to sunlight, especially after several pregnancies

Vit D toxicity

1-Results from excessive supplementation –long term therapeutic doses

2-Some infants are sensitive to intakes of 50 µg/day

leads to hypercalcimia and hyperphosphatemia

and leads to contraction of blood vessels: high blood pressure, and (calcinosis) is calcium deposit in soft tissues(kidney, myocardium) and synovial membranes

- Over reabsorption of calcium (hypercalcemia) may develop kidney stones

- Excessive exposure to sun light does not lead to vit D toxicity because there is

1) a limited capacity to form the precursor 7 –dehydrocholesterol and

2) a limited capacity to take up cholecalciferol from the skin

- While increased sunlight exposure would meet the need, it carries the risk of developing skin cancer.